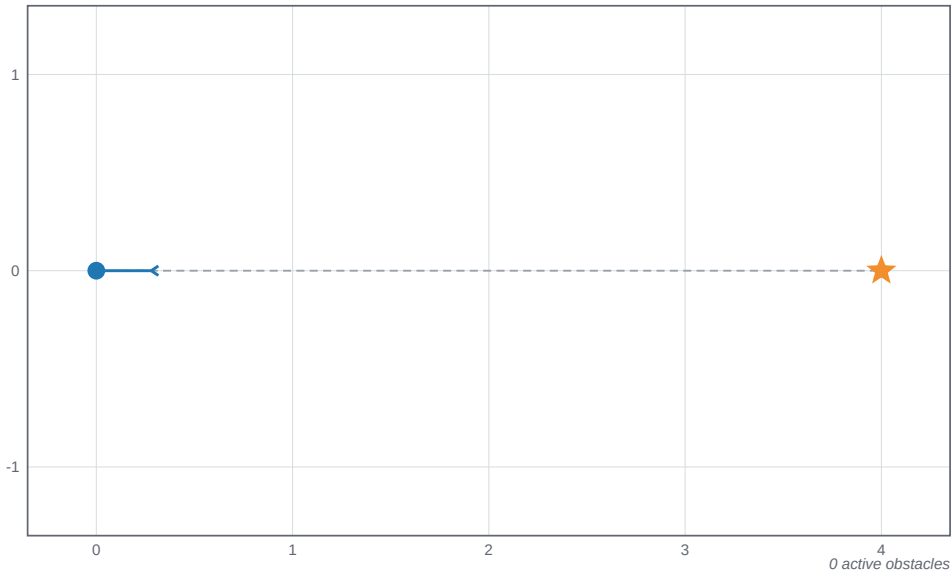


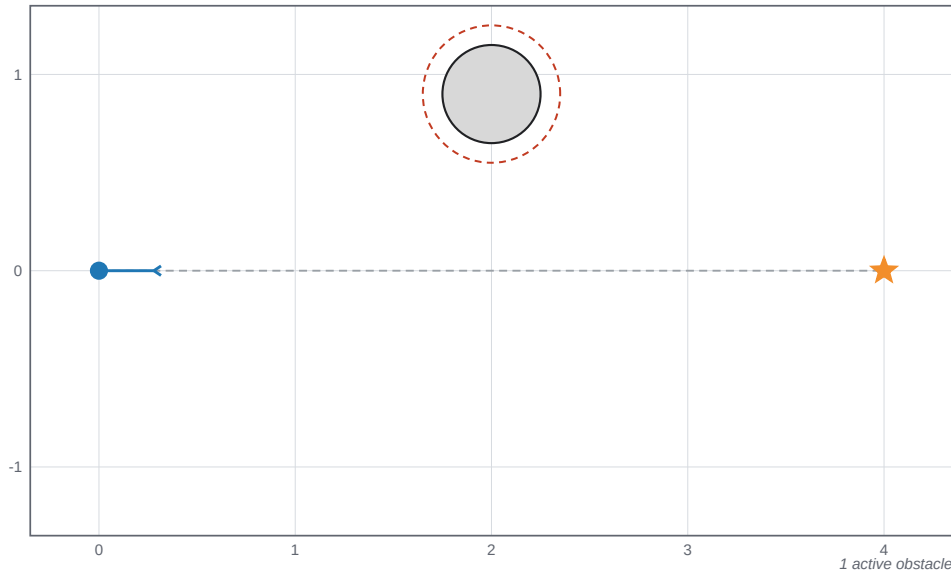
Control and noninterference layouts

Geometrically easy cases used to check basic navigation and unnecessary intervention.

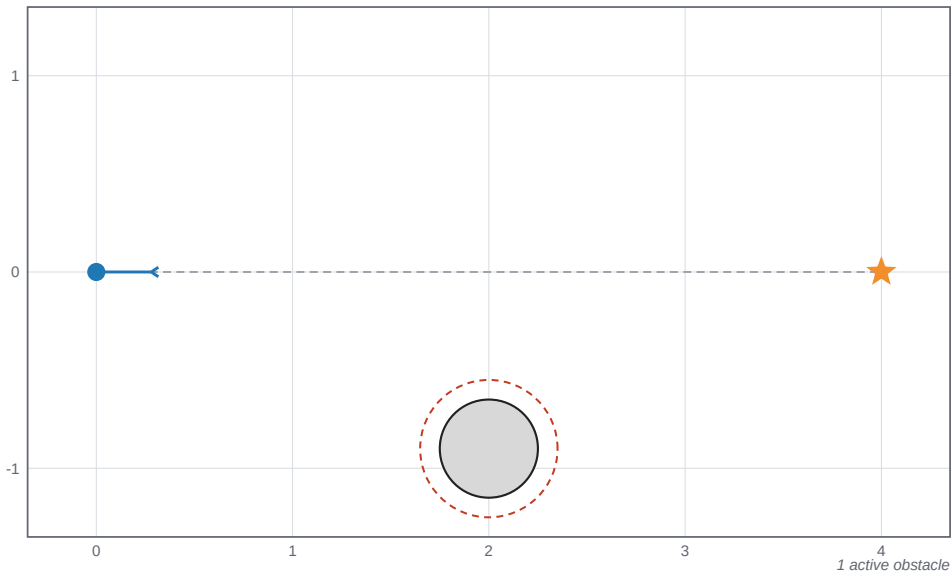
control open route



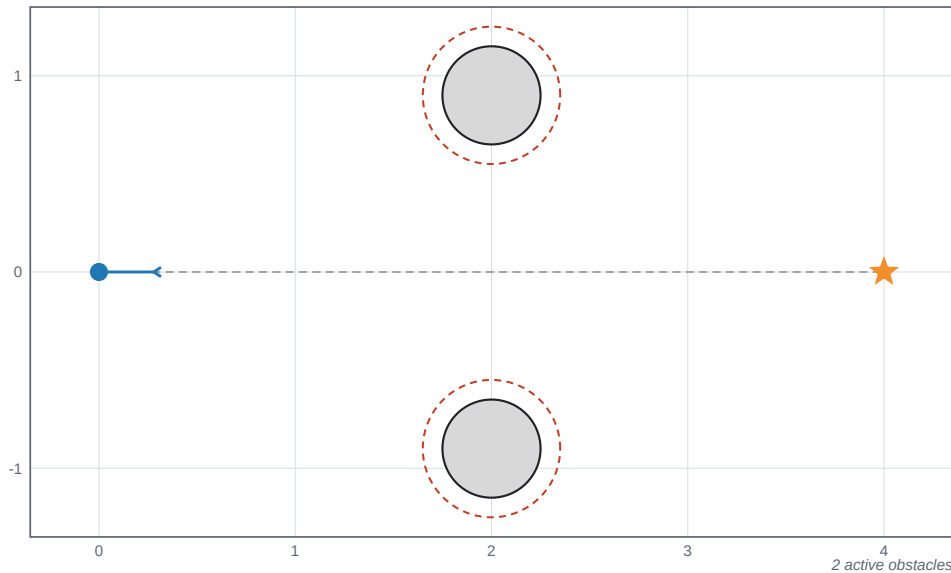
control upper clearance



control lower clearance



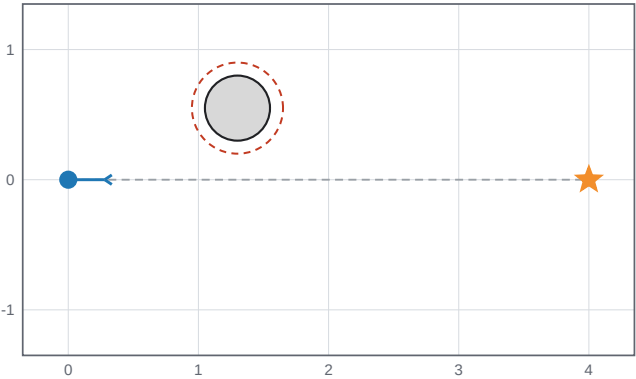
control symmetric clearance



Single-obstacle deviations

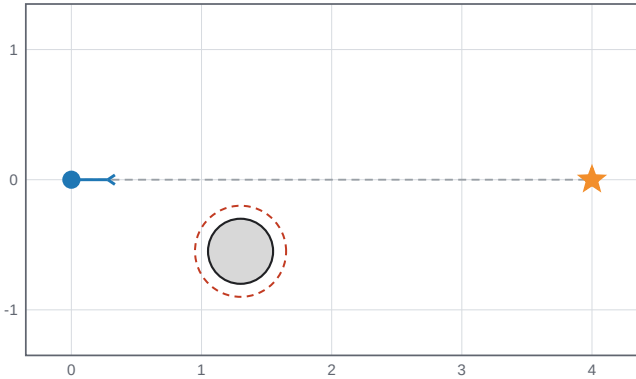
One obstacle varies in longitudinal position, lateral offset, and direct-path obstruction.

single near early upper



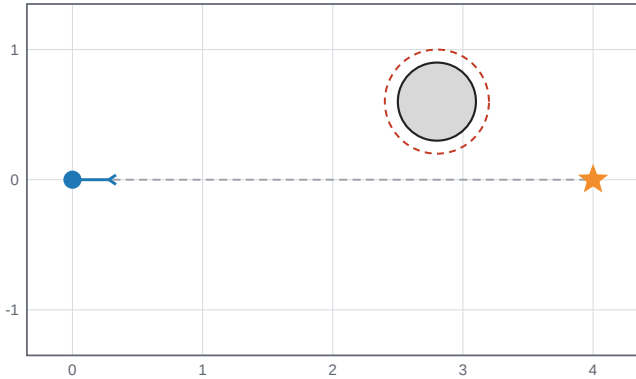
1 active obstacle

single near early lower



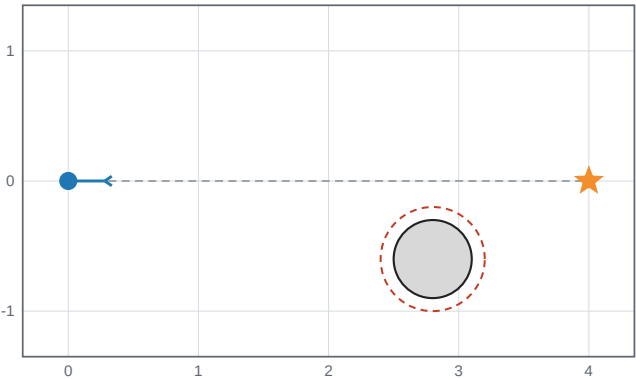
1 active obstacle

single near late upper



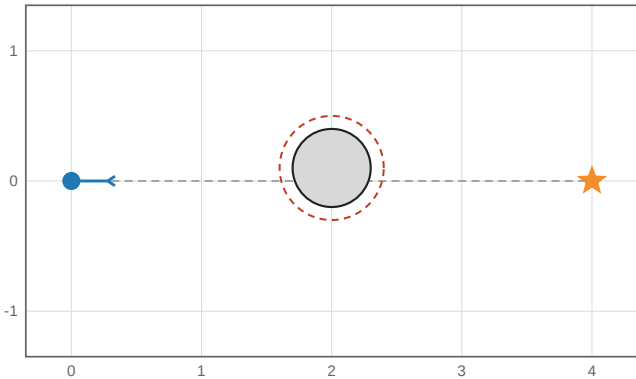
1 active obstacle

single near late lower



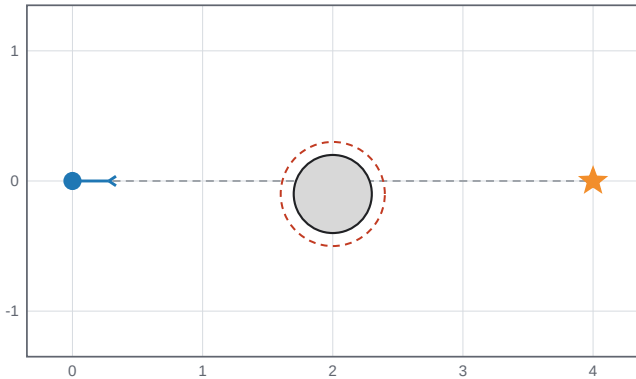
1 active obstacle

single blocked central upper



1 active obstacle

single blocked central lower



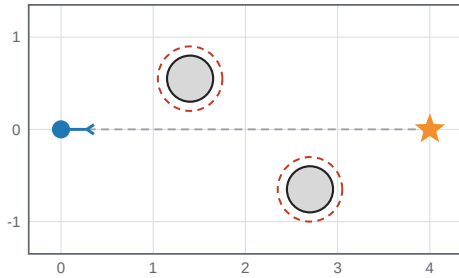
1 active obstacle

Two-obstacle interactions

8 layouts | suite: core_navigation_layouts_v1

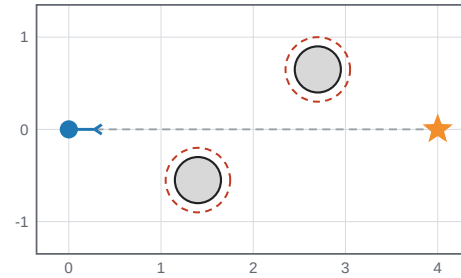
Two obstacles create staggered, same-side, near-path, and direct-path-blocking cases.

double near staggered upper
first



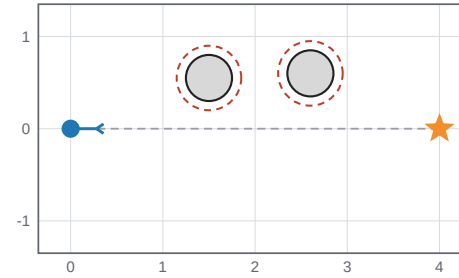
2 active obstacles

double near staggered lower
first



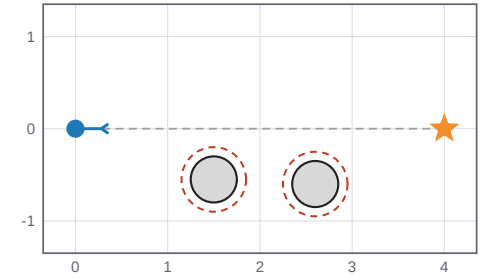
2 active obstacles

double near same side upper



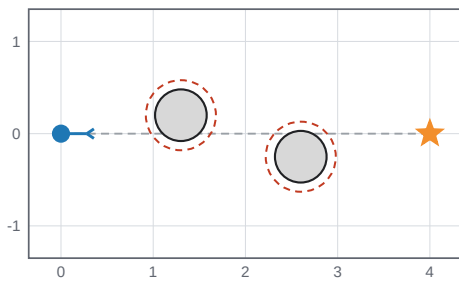
2 active obstacles

double near same side lower



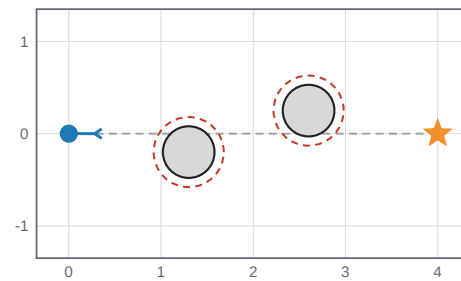
2 active obstacles

double blocked staggered upper
first



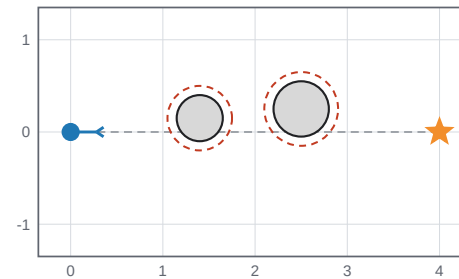
2 active obstacles

double blocked staggered lower
first



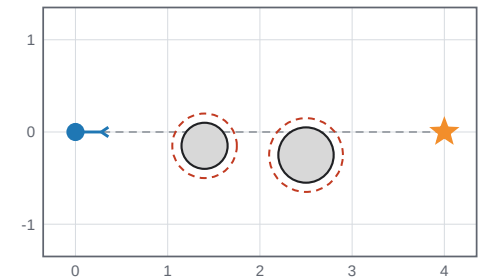
2 active obstacles

double blocked same side upper



2 active obstacles

double blocked same side lower



2 active obstacles

● start and initial heading

★ goal

○ physical obstacle

○ collision boundary (obstacle + agent radius)

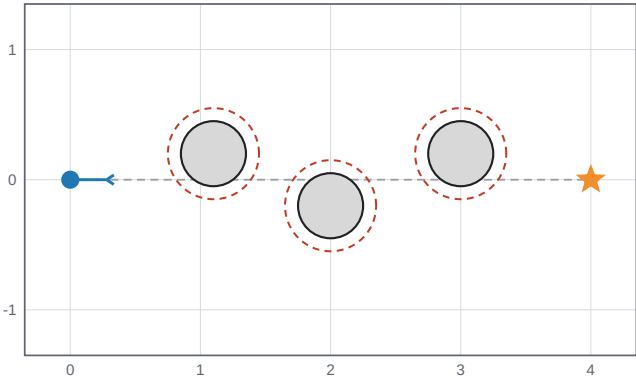
--- straight start-goal reference

All layouts: start=(0,0), heading=0, goal=(4,0), agent radius=0.10

Three-obstacle slaloms

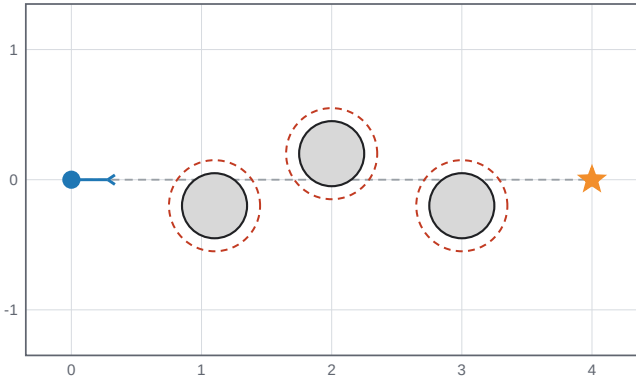
Alternating three-obstacle patterns provide the most constrained geometric cases.

triple mild slalom upper first



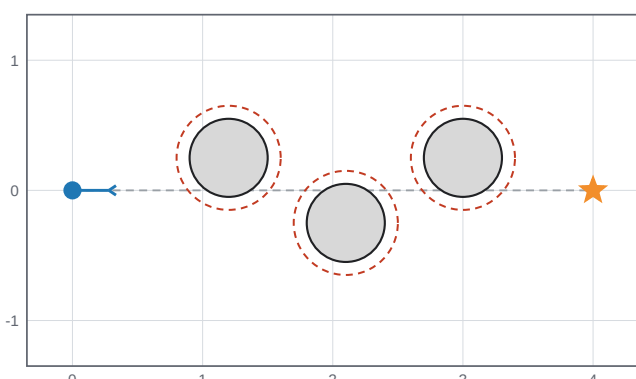
3 active obstacles

triple mild slalom lower first



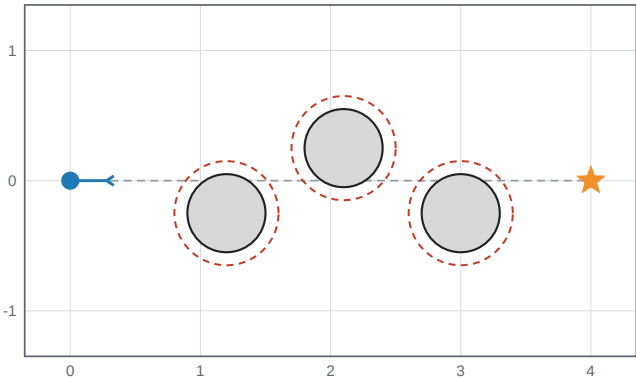
3 active obstacles

triple narrowing slalom upper first



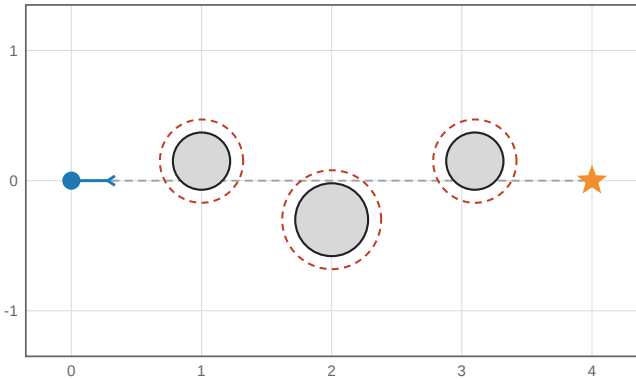
3 active obstacles

triple narrowing slalom lower first



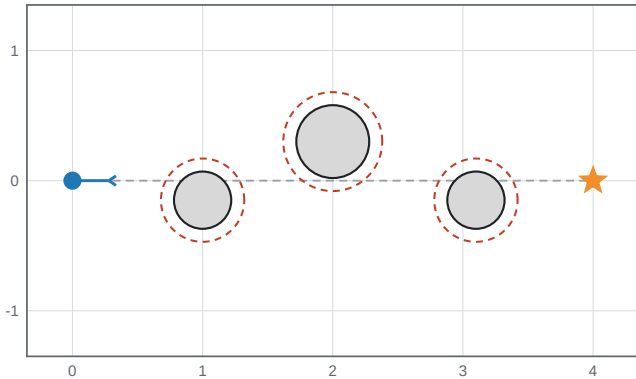
3 active obstacles

triple shifted slalom upper first



3 active obstacles

triple shifted slalom lower first



3 active obstacles